Fiscal Year:	FY 2008	**Task Last Updated:**	FY 04/29/2008
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sdytse@rutgers.edu	**Fax:**	FY 609 258-6233
PI Organization Type:	UNIVERSITY	**Phone:**	609 258-5271
Organization Name:	Princeton University		
PI Address 1:	Department of Mechanical and Aerospace Engineering		
PI Address 2:			
PI Web Page:			
City:	Princeton	**State:**	NJ
Zip Code:	08544	**Congressional District:**	12
Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	12/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Sacksteder, Kurt R	**Contact Phone:**	216-433-2857
Contact Email:	Kurt.R.Sacksteder@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: this grant on NCE until 12/31/2007, per K. Sacksteder/GRC (11/07) NOTE: grant to be reinstated with a new grant number. 8/2006, per D. Urban		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Tse, Stephen (Rutgers University) Sacksteder, Kurt (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1051		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	The transient behavior of burner-supported spherical diffusion flames was studied in the transport-induced limit of low mass flow rate and the radiation-induced limit of high mass flow rate which characterize the isola response of flame extinction. Oscillatory instability was observed near both steady-state extinction limits. The oscillation typically grows in amplitude until it becomes large enough to extinguish the flame. The oscillatory behavior was numerically observed using detailed chemistry and transport for methane (50%CH4/50%He into 21%O2/79%He) and hydrogen (100% H2 into 21%O2/79%He) diffusion flames where the fuel was issued from a point source, and helium was selected as an inert to increase the Lewis number, facilitating the onset of oscillation. In both methane and hydrogen flames, the oscillation always leads to extinction, and no limit cycle behavior was found. The growth rate of the oscillation was found to be slow enough under certain conditions to allow the flame to oscillate for over 450 seconds, suggesting that such oscillations can possibly be observed experimentally. For the hydrogen flames, however, the frequency of oscillation near the transport-induced limit is much larger, approximately 60 Hz as compared to 0.35 Hz for the methane flame, and the maximum amplitude of temperature oscillations was about 5K. The distinctively different structures of the hydrogen and methane flames suggest that while both instabilities are thermal-diffusive in origin, oscillations in the hydrogen flames resemble those of premixed flames, while oscillations in the methane flames are non-premixed in character. The structure and extinction characteristics of counterflow diffusion flames with flame radiation and non-unity Lewis numbers of the fuel and oxidant were examined using multi-scale asymptotic theory and a model expressed in terms of the jump relations and reactant leakages with the proper consideration of the excess enthalpy overlooked in previous analyses is developed. The existence of the dual extinction limits in the presence of radiative heat loss, namely the kinetic limit at small Damkohler number (high stretch rate) and the radiative limit at large Damkohler number (low stretch rate), are identified. It is found that the former one is minimally affected by radiative loss, while substantial amount of heat loss is associated with the radiative limit. The influence of radiative loss on the extinction Damkohler numbers is found to be through its effects on the flame temperature, excess enthalpy and the reduced extinction Damkohler number. At both extinction limits, the contribution from the flame temperature is always important and dominant. The contributions from the other two, however, could be important in some special cases. At small Le (fuel), the contribution from the reduced extinction Damkohler number is large and even dominant under small radiative loss. The contribution from the excess enthalpy is important for small Le (oxidizer) and it may be comparable to the contribution from the flame temperature when radiative loss is small. Thus, overlooking the excess enthalpy by previous analyses may result in rather large error in the predicted extinction Damkohler numbers, especially the kinetic one. Using the results derived from the previous study, a linear stability analysis was conducted to study the onset of near-limit flame oscillation with radiative heat loss in 1-D chambered planar flames using multi-scale activation-energy asymptotics. The oscillatory instability near the radiation-induced extinction limit at large Damkohler numbers is identified, in additional to the one near the kinetic limit at small Damkohler numbers. It is shown that radiative loss assumes a similar role as varying the thermal diffusivity of the reactants. Thus, flame oscillation near the radiative limit is still thermal-diffusive in nature although it may develop under unity Lewis numbers. The unstable range of Damkohler numbers near the radiative limit shows quite similar parametric dependence on the Lewis numbers of reactants, Le (fuel) and Le (oxidizer), the equivalence ratio, and the radiative loss as that near the kinetic limit. They both increase monotonically with Le (oxidizer) and equivalence ratio and increase then decrease with Le (fuel). Increasing radiative loss extends the parameter range under which flame oscillations may develop. However, they show different dependence on the temperature difference between the supplying reactants. Unless radiative loss approaches its maximum value the system can sustain, flame oscillation near the radiative limit is only possible within a limited range of delta-T, whereas it is promoted monotonically with decreasing delta-T near the kinetic limit. Furthermore, while radiative loss shows small effect on the nondimensional oscillation frequency, the dimensional frequency of flame oscillations near the radiative limit can be substantially smaller than that near the kinetic limit. We have performed microgravity experiments on the effects of rotation on spherical diffusion flames, with emphasis on varying Lewis numbers on both the fuel and oxidizer sides. The study was motivated by recent theoretical development on the effects of rotation on droplet- or burner-generated flames. The experiments were devised not only to verify the theoretical predictions, but also extend or supplement the theories to study the flame response at high angular velocities where the perturbation theory no longer applies. The experiments were conducted using the 2.2 second drop tower at GRC. The flames were generated by injecting either the fuel or the oxidizer mixtures through a bronze porous burner to a controlled ambient which also comprises either an oxidizer or fuel mixture. As the theory predicted, the flame strengthened either at the pole or the equator as the Lewis numbers of the ambient and burner mixtures were varied.

	Specifically, it was demonstrated that when subjected to rotation, the flame deformed into a pancake shape, flattened at the poles and protruding outward from the equator, regardless of Le (fuel) and Le (oxidizer). The deformation increased as the angular velocity increased but reached an asymptote at high angular velocities, at which the flame surface approached very close to the burner surface near the poles. The flame temperature, however, was noticeably influenced by the Le of the system. When the Le of the burner mixture was less than unity, the emission at the poles significantly became stronger as rotation increased, even though heat loss to the burner was increasing as the flame became more flat. When this fuel mixture was moved to the ambient, just the opposite was observed. In fact, if these spherical flames were rendered to be very weak, the current results indicate that rotation could give rise to local extinction either at the poles or the equator. Finally, the flame shape deformations were shown to increase linearly with mass flow rates. All these observations were in accordance with the results of our theoretical work, to be discussed next. The configuration and extinction of diffusion flames generated by either a porous spherical burner or a fuel droplet in response to slow rotational motion were investigated through perturbation analysis, with emphasis on the effects of non-unity Lewis numbers for both fuel and oxidizer. It is shown that the rotational motion induces a secondary flow which is directed inward towards the poles and outward from the equator, which in turn distorts the otherwise spherical flame into a pancake shape that is flattened at the poles and protruding at the equator. The flame temperature is also affected, becoming more susceptible to extinction either at the poles or the equator, depending on the combined effects of Lewis numbers, flame stretch through local flow nonuniformity, and ambient oxidizer concentration. The differences and similarities between the responses of diffusion and premixed flames in response to rotation are discussed. In the previous work, by assuming constant density a perturbation analysis was carried out to examine effects of the secondary flow induced by rotation on the response of spherical diffusion flames enveloping either a burner or a fuel droplet. Complete analytical solutions were found for all the relevant quantities, with explicit expressions derived for the burning rate and distorted flame shape. General Lewis numbers were considered, as were ambient oxidizer concentration. The effect of rotation on local extinction was also assessed.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
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